

A Uniform Ext Constant for Normalized Exact Sequences

arXiv automatic math run `fa_banach_001`

Candidate packet, June 14, 2026

Classification

- **Status:** conditional / normalized exact-sequence result.
- **Source:** P. Kochanek, *Stability of vector measures and twisted sums of Banach spaces*, arXiv:1208.4755.
- **Target:** final Problem (6), asking whether the property $\text{Ext}(\mathfrak{X}, \cdot) = 0$ is a uniform three-space property.

Source Question

The source paper asks whether there is a function

$$\varphi : (0, \infty)^2 \longrightarrow (0, \infty)$$

such that, whenever $\mathfrak{X}, X, Y, Z$ are Banach spaces, $0 \rightarrow Y \rightarrow Z \rightarrow X \rightarrow 0$ is exact, and $\text{Ext}(\mathfrak{X}, X) = 0 = \text{Ext}(\mathfrak{X}, Y)$, then

$$\text{Ext}(\mathfrak{X}, Z) = 0 \quad \text{and} \quad Z(\mathfrak{X}, Z) \leq \varphi(Z(\mathfrak{X}, X), Z(\mathfrak{X}, Y)).$$

Here $Z(U, V)$ is the best constant for approximating zero-linear maps $U \rightarrow V$ by linear maps, as in the source paper.

view of Theorem 8.2, a positive answer (even under the stronger assumption) would imply that $\text{Ext}(c_0, Y) \neq 0$ for every infinite-dimensional, reflexive Banach space Y , *i.e.* every such space would produce a non-trivial locally convex twisted sum with c_0 . Cabello Sánchez and Castillo showed that $\text{Ext}(c_0, Y) \neq 0$ whenever Y is a cotype 2 space, complemented in its bidual ([9, Corollary 1]).

- (5) A weaker variant of (2) and (3): Does the **SVM** property of X imply that X^{**} is injective?
- (6) The assumption $\text{cf}(\kappa) > \omega$ was made in Theorem 7.1, because although we know that the property $\text{Ext}(\mathfrak{X}, \cdot) = 0$ is a **3SP** property, we do not know if it is a **3SP** property ‘uniformly’. More precisely, the question reads as follows: Is there a function $\varphi : (0, \infty)^2 \rightarrow (0, \infty)$ such that whenever $\mathfrak{X}, X, Y, Z$ are Banach spaces, $0 \rightarrow Y \rightarrow$

$Z \rightarrow X \rightarrow 0$ is an exact sequence, and $\text{Ext}(\mathfrak{X}, X) = 0 = \text{Ext}(\mathfrak{X}, Y)$, then

$$\text{Ext}(\mathfrak{X}, Z) = 0 \quad \text{and} \quad Z(\mathfrak{X}, Z) \leq \varphi(Z(\mathfrak{X}, X), Z(\mathfrak{X}, Y))?$$

If the answer is ‘yes’, then we may drop the assumption $\text{cf}(\kappa) > \omega$ in Theorem 7.1, by using Corollary 7.6. Note that the homological proof of the fact that $\text{Ext}(\mathfrak{X}, \cdot) = 0$ is

Conditional Dependencies

The theorem below is proved for exact sequences in the following normalized form. We assume that

$$0 \longrightarrow Y \xrightarrow{j} Z \xrightarrow{q} X \longrightarrow 0$$

has isometric kernel inclusion and admits a homogeneous right inverse $b : X \rightarrow Z$ of q such that

$$qb = \text{id}_X, \quad \|b(x)\| \leq s\|x\| \quad (x \in X).$$

This is automatic, for every $s > 1$, in the canonical metric-quotient model where Y is a closed subspace of Z , $X = Z/Y$ has the quotient norm, and q is the quotient map.

Thus, if the source question is interpreted in this normalized model, the corollary gives an affirmative bound. If the question is interpreted literally for arbitrary endpoint identifications in exact sequences, this packet leaves one dependency: the quotient open-map and embedding distortion constants would also have to be controlled, or shown irrelevant, using only $Z(\mathfrak{X}, X)$ and $Z(\mathfrak{X}, Y)$.

Proof Intuition

Let $F : E \rightarrow Z$ be zero-linear. Push F down to the quotient X and use $Z(E, X)$ to straighten qF by a linear map. Lift that linear map algebraically back to Z ; boundedness is not needed because linear approximants in the zero-linear definition are not required to be continuous. After subtracting the lift, the remaining map has small quotient part. A homogeneous selector lifts this small quotient part back into Z ; subtracting it leaves a zero-linear map into the kernel Y . Applying $Z(E, Y)$ to this kernel-valued map finishes the approximation. The only constant loss comes from lifting the quotient error through the selector.

Theorem

Theorem 1. *Let E, X, Y, Z be Banach spaces and suppose that*

$$0 \longrightarrow Y \xrightarrow{j} Z \xrightarrow{q} X \longrightarrow 0$$

is exact, j is an isometry onto $\ker q$, $\|q\| \leq 1$, and q has a homogeneous right inverse $b : X \rightarrow Z$ satisfying

$$qb = \text{id}_X, \quad \|b(x)\| \leq s\|x\| \quad (x \in X).$$

Put $A = Z(E, X)$ and $B = Z(E, Y)$, allowing A or B to be $+\infty$. If $A, B < \infty$, then

$$Z(E, Z) \leq sA + B(1 + 2sA).$$

Proof. Let $F : E \rightarrow Z$ be zero-linear and write $\delta = Z(F)$. If $\delta = 0$, the same argument below applies with every estimate equal to 0, so assume $\delta > 0$.

Since $\|q\| \leq 1$, the composition $qF : E \rightarrow X$ is zero-linear and $Z(qF) \leq \delta$. By the definition of A , for every $\varepsilon > 0$ there is a linear map $\ell : E \rightarrow X$ such that

$$\text{dist}(qF, \ell) \leq (A + \varepsilon)\delta.$$

Choose an algebraic linear lift $L : E \rightarrow Z$ of ℓ , so that $qL = \ell$. No boundedness of L is required. Set

$$G = F - L, \quad u = qG = qF - \ell.$$

Then G is zero-linear with $Z(G) = \delta$, and by homogeneity

$$\|u(x)\| \leq (A + \varepsilon)\delta\|x\| \quad (x \in E).$$

Define

$$H = G - bu.$$

Because $qb = \text{id}_X$, we have $qH = 0$, hence $H(E) \subseteq j(Y)$. Identifying Y with $j(Y)$, we estimate the zero-linearity of H . For any finite family $x_1, \dots, x_n \in E$,

$$\begin{aligned} & \left\| H \left(\sum_i x_i \right) - \sum_i H(x_i) \right\| \\ & \leq \left\| G \left(\sum_i x_i \right) - \sum_i G(x_i) \right\| + \left\| bu \left(\sum_i x_i \right) - \sum_i bu(x_i) \right\| \\ & \leq \delta \sum_i \|x_i\| + s \left\| u \left(\sum_i x_i \right) \right\| + s \sum_i \|u(x_i)\| \\ & \leq (1 + 2s(A + \varepsilon))\delta \sum_i \|x_i\|. \end{aligned}$$

Thus H is a Y -valued zero-linear map with

$$Z(H) \leq (1 + 2s(A + \varepsilon))\delta.$$

By the definition of B , for every $\eta > 0$ there is a linear $m : E \rightarrow Y$ satisfying

$$\text{dist}(H, jm) \leq (B + \eta)(1 + 2s(A + \varepsilon))\delta.$$

Then $L + jm : E \rightarrow Z$ is linear, and for $\|x\| \leq 1$,

$$\begin{aligned} \|F(x) - (L + jm)(x)\| &= \|G(x) - jm(x)\| \\ &\leq \|H(x) - jm(x)\| + \|bu(x)\| \\ &\leq (B + \eta)(1 + 2s(A + \varepsilon))\delta + s(A + \varepsilon)\delta. \end{aligned}$$

Taking the infimum over admissible linear approximants and then letting $\varepsilon, \eta \downarrow 0$ gives

$$Z(E, Z) \leq sA + B(1 + 2sA).$$

□

Corollary 1. *In the canonical metric-quotient normalization $0 \rightarrow Y \rightarrow Z \rightarrow Z/Y \rightarrow 0$,*

$$Z(E, Z) \leq Z(E, Z/Y) + Z(E, Y) + 2Z(E, Z/Y)Z(E, Y).$$

Consequently, for a normalized exact sequence $0 \rightarrow Y \rightarrow Z \rightarrow X \rightarrow 0$ with X carrying the quotient norm,

$$\varphi(a, b) = a + b + 2ab$$

is an admissible uniform three-space bound.

Proof. For each $s > 1$, the quotient map admits a homogeneous selection $b : X \rightarrow Z$ with $\|b(x)\| \leq s\|x\|$: choose, for each non-zero ray in X , a representative within the factor s of the quotient norm and extend homogeneously. Apply the theorem and let $s \downarrow 1$. $\square$

Verification and Novelty Notes

The proof uses only the definitions of zero-linear maps, the stability constant $Z(E, \cdot)$, and the standard homogeneous selection for a quotient map. The linear lift L is purely algebraic, which is legitimate because the linear maps used in the definition of $Z(E, V)$ need not be bounded.

The bounded novelty check for this packet searched the run registry and solution/attempt indexes for arXiv:1208.4755 and for the core phrases “uniform 3SP”, “zero-linear”, and “ $Z(E, Z)$ ”. No existing packet or attempt for this exact target was found. Short phrase searches for the displayed bound and the source problem did not reveal an obvious literature duplicate. The result is therefore recorded as a candidate new conditional/normalized estimate, not as a claimed full answer under every possible interpretation of the source question.

Human Review Recommendation

Reviewers should first decide whether Kochanek’s final question is intended only up to the canonical quotient normalization. If yes, the corollary appears to give a direct affirmative answer with

$$\varphi(a, b) = a + b + 2ab.$$

If arbitrary endpoint identifications are part of the intended formulation, the reviewer should test whether the missing quotient and embedding distortion constants can be absorbed by the endpoint stability constants alone.

References

- [1] P. Kochanek, *Stability of vector measures and twisted sums of Banach spaces*, arXiv:1208.4755.
- [2] F. Cabello Sanchez and J. M. F. Castillo, *Stability constants and the homology of quasi-Banach spaces*, Israel J. Math. 198 (2013), 347–370.